

Sensors and Actuators: A. Physical

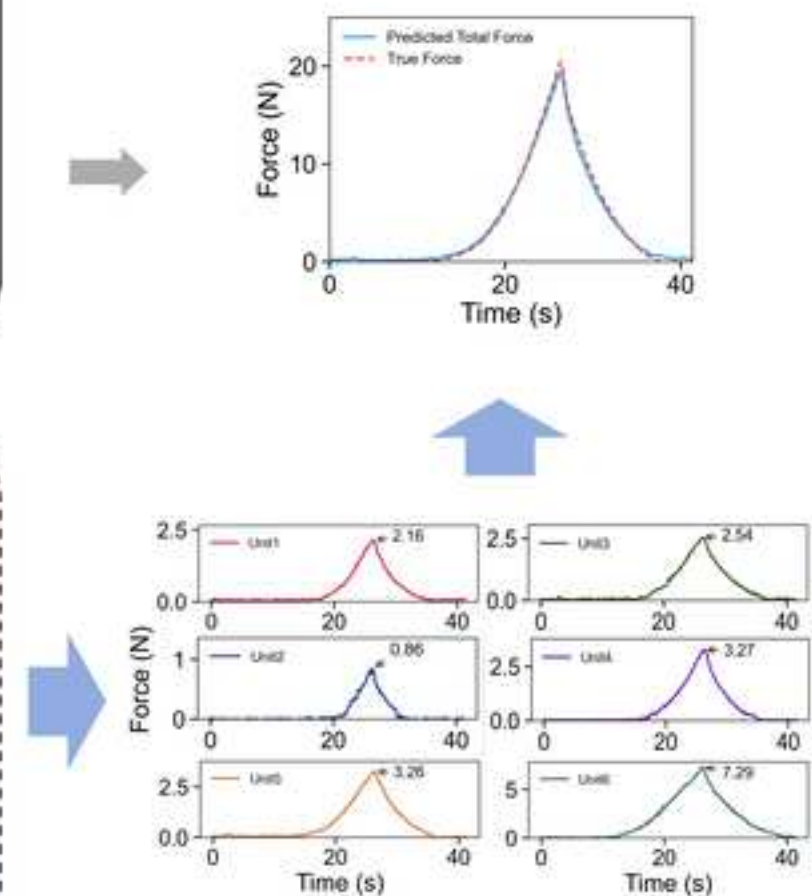
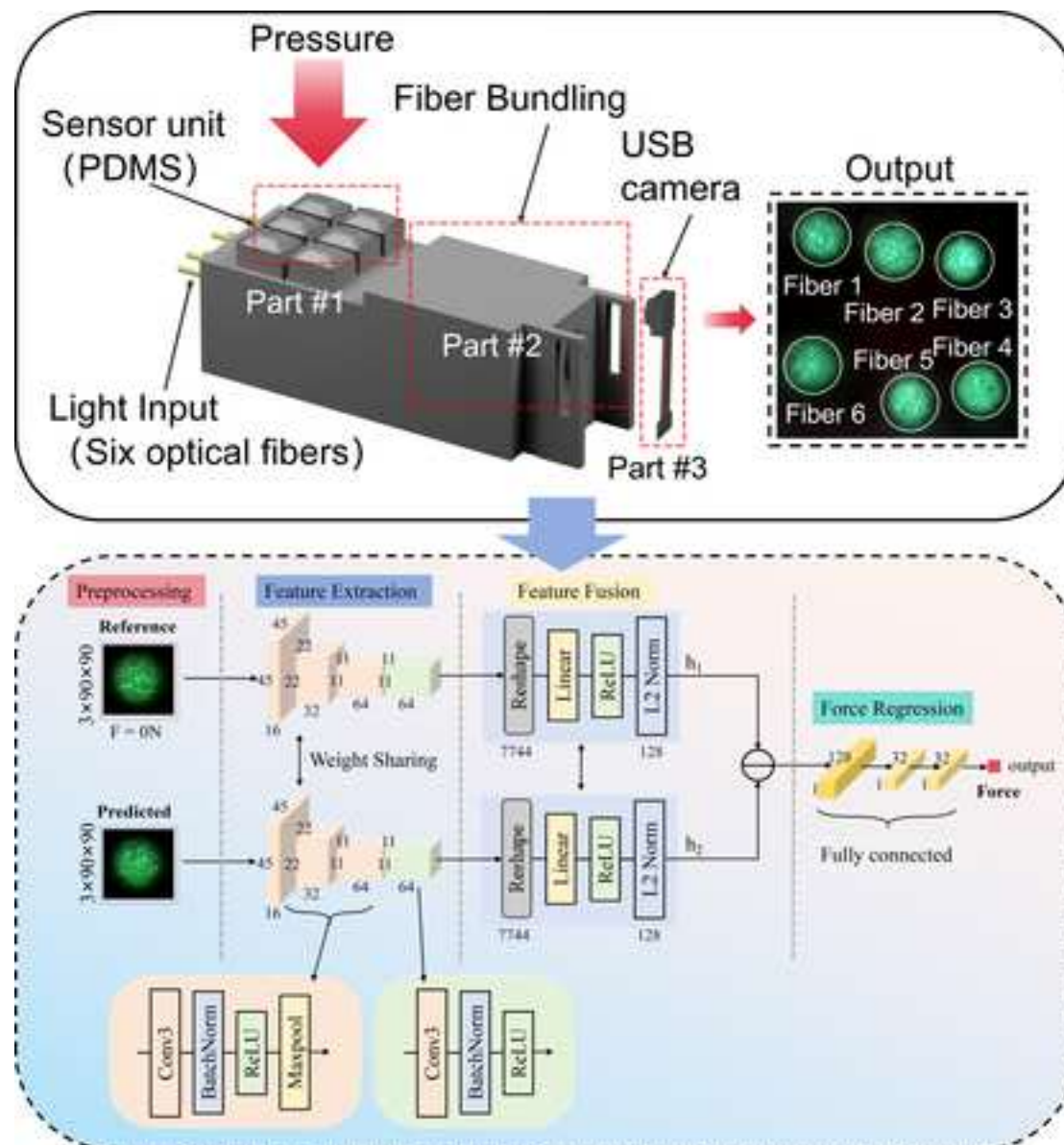
Optical Fiber Tactile Sensing Array Based on Dynamic Learning of Optical Specklegram

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Highlights:

Low-cost, miniaturized Multi-MMF array enables multi-point force sensing.

Proposed DDL-Net achieves high generalization by analyzing specklegram differences.

Dynamic reference specklegram updating strategy realizes sensor zeroing to enhance stability.

High accuracy under different light intensities leads to good usability.

Optical Fiber Tactile Sensing Array Based on Dynamic Learning of Optical Specklegram

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1. Introduction

Humans inherently possess tactile perception, which relies on skin-based receptors. These receptors convert mechanical stimuli generated by finger contact into electrical signals, which are then transmitted to the brain. This sensory feedback mechanism enables the brain to regulate finger joint movements and adjust grip force accordingly, thereby ensuring the safe and stable grasping of objects [1]. Similarly, in industrial settings,

tactile sensors deliver real-time feedback throughout the robotic grasping process, thereby optimizing task efficiency. Specifically, they enable precise regulation of grip force: this not only prevents objects from slipping, dropping, or sustaining damage but also strengthens the stability of robot-object interactions. Owing to these capabilities, such tactile sensors exhibit broad application potential in industrial scenarios [1-4]. Currently, various tactile sensors are widely used in fields such as medicine, agriculture, and industry. Different application scenarios impose distinct requirements on sensor design. However, some issues persist as prominent challenges in the practical deployment of such tactile sensors, including nonlinear responses, poor repeatability, susceptibility to noise interference, significant hysteresis, and vulnerability to shear forces, and the difficulty of integrating complex sensing systems into the limited space of robotic dexterous hands [5].

Tactile sensing can be implemented through multiple approaches, including capacitive [6-7], piezoelectric [8], piezoresistive [9], visual-tactile fusion [11], and fiber-optic techniques [12-14]. Among these, fiber-optic sensors stand out due to their high precision, fast response, and high anti-interference capabilities. Furthermore, they can be fabricated into various shapes, offering flexibility to adapt to diverse application scenarios [15-16]. Although distributed sensing technologies like Fiber Bragg Grating (FBG) offer high spatial resolution along a single fiber, they often require bulky and expensive spectral interrogators, limiting their suitability for cost-sensitive and compact robotic applications. Fiber Specklegram Sensing (FSS), as an emerging fiber-optic sensing technology, retains the merits of traditional optical sensors while also offering a simplified demodulation process and reduced cost [17]. FSS operates by analyzing the specklegram patterns formed at the output end face of multimode fibers and has been successfully utilized for measuring various physical quantities [18-21]. These specklegram patterns, resulting from modal interference within the multimode fiber, appear as a non-uniform distribution of light and dark intensities at the output end face. In contrast to intensity-based modulation, which relies on macroscopic bending loss, FSS leverages Interference between modes to provide exceptional sensitivity to minute perturbations. However, this inherent high sensitivity also makes FSS susceptible to environmental noise and light intensity fluctuations, leading to specklegram drift [22-23]. Various analysis methods have been employed for FSS, including statistical image analysis [24] and deep learning techniques [25-26]. However, most of the FSS tactile sensor research remains in its early stages, with few studies exploring their practical viability in real-world settings. Current efforts to enhance FSS stability focus on two main strategies: (1) optimizing fiber structure to reduce environmental impact [27], for instance by shortening multimode fibers to lower environmental sensitivity; and (2) improving sensing algorithms to suppress disturbances. For deep learning-based FSS sensors, model robustness can be improved by expanding training datasets (e.g., collecting data across different time periods [28]) or using transfer learning with new data when performance declines [29]. However, the necessity for frequent data collection complicates calibration and impedes deployment, limiting its practical utility.

To enhance the practical viability of FSS under real-world constraints, this study improves both hardware design and algorithms. In terms of hardware, we present a compact integrated, miniaturized fiber array sensor featuring a USB camera and an custom-designed geometric housing. This structure isolates six optical fibers into distinct compartments to mechanically decouple sensing regions, while the compact integration notably shortens the fibers to minimize environmental noise. On the algorithmic front, we examine the Zero-Mean Normalized Cross-Correlation (ZNCC) method and, drawing on its dual-image comparison logic, propose a Dynamic Difference Learning (DDL) framework incorporating CNNs. Experimental results show that difference-based learning is effective for specklegram sensing, and that the DDL framework effectively mitigates signal fluctuations by dynamically updating reference specklegrams to achieve robust performance. Specifically, when analyzing speckle patterns collected 24 hours after model training, DDL yields a lower RMSE than single-specklegram image learning with negligible computational overhead. Both the sensor and

algorithm designs are compatible with real-time sensing requirements and allow for the perception of force magnitude, distribution, and resultant force.

2. Sensor Design

2.1 Working Principles

When coherent light is transmitted from a single-mode fiber (SMF) to a multi-mode fiber (MMF), the MMF supports the propagation of multiple modes. The interference between these modes generates a non-uniform intensity distribution, forming a complex speckle pattern known as a specklegram. By simplifying the light field propagation, the complex amplitude across the fiber output end face is typically formulated as Eq. (1) [30]:

$$U(x, y) = \sum_{m=0}^{M-1} a_{0m}(x, y) \exp(j\varphi_{0m}(x, y)) \quad (1)$$

Where M denotes the total number of propagating modes in the fiber, a_{0m} and φ_{0m} represent the amplitude and phase of the m -th mode, respectively.

If the light field is captured by a camera positioned at the output end of the fiber, the light intensity mapped to each point on the camera plane can be expressed by Eq. (2):

$$I_0(x, y) = |U_0(x, y)|^2 = \sum_{m=0}^{M-1} \sum_{n=0}^{M-1} a_{0m}(x, y) a_{0n}(x, y) \exp(j[\varphi_{0m}(x, y) - \varphi_{0n}(x, y)]) \quad (2)$$

From Eq. (2), it can be seen that the complexity of specklegrams stems from the mode interference. As the number of excited modes M increases, Eq. (2) contains more terms, leading to a more intricate light intensity distribution.

When external force is applied to the optical fiber, the elasto-optic effect modifies the fiber's refractive index distribution, while material deformation alters its length. These changes ultimately affect the effect of mode interference. Consequently, variations in the output specklegram pattern of the MMF can be utilized to detect external force changes.

2.2 Sensor Design and Fabrication

The sensor design targets three primary objectives: miniaturization and integrability, with each sensing unit measuring $5 \times 7 \text{ mm}^2$; robust sensing performance, characterized by high sensitivity and repeatability; and an independent array architecture, where physically isolated slots and dedicated fibers eliminate crosstalk to enable precise force distribution mapping.

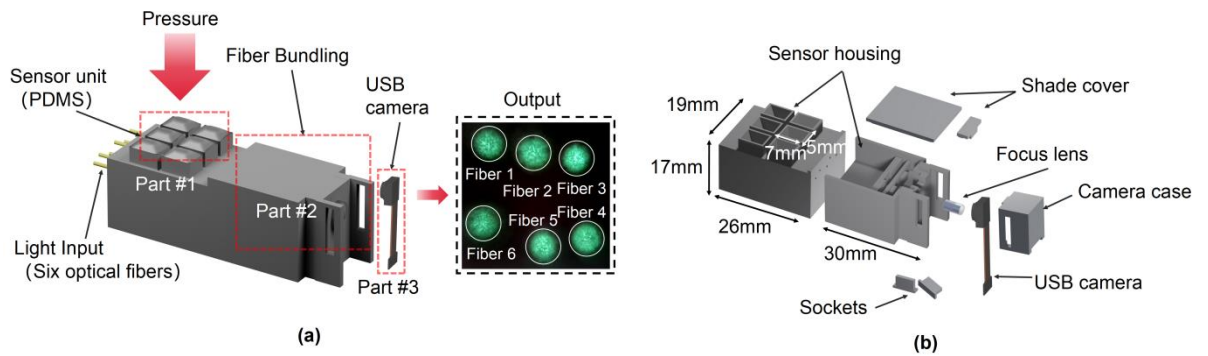


Fig. 1. Sensor design and implementation: (a) Working principle schematic. (b) Exploded view of sensor components.

As shown in Fig. 1(a), the sensor adopts a modular design, divided into three parts: Part 1 is the sensing area, composed of a 2×3 array of sensing units, each utilizing a SMF-MMF fusion structure; Part 2 is the fiber optic convergence region. The six MMF output ends are bent and converge into the micro-lens, through which the output speckle patterns are focused onto the imaging plane without overlap; Part 3 is the signal acquisition module, consisting of a USB camera that captures real-time specklegram changes. This configuration allows for shortened multimode fibers, effectively mitigating environmental noise associated with long-distance transmission.

Fig. 1(b) shows an exploded view of the sensor's core structure, which includes a sensor housing, a fiber optic focusing lens, and a micro USB camera module. The sensor housing is fabricated using fused deposition modeling (FDM) 3D printing with polylactic acid (PLA). A socket design secures the USB camera to the housing, forming an integrated assembly. The lens, embedded within the converging structure, is located directly in front of the camera's imaging plane.

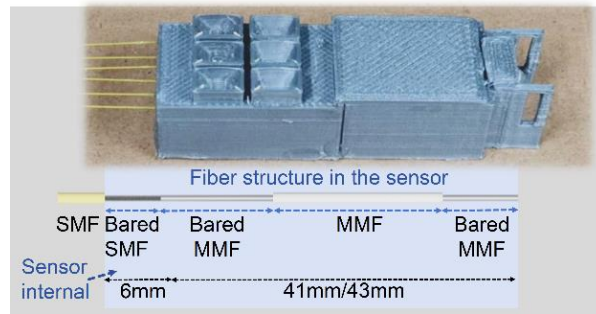


Fig. 2. Physical photograph of the fabricated sensor and optical fiber structure in the sensor.

The SMF-MMF fiber structure is used in each unit as shown in Fig. 2. The SMF at the input retains its coating for tensile strength, while the MMF in the sensing region is stripped of its coating to improve sensitivity. The MMF in the bundling region retains its coating to prevent bending fractures, ensuring overall durability.

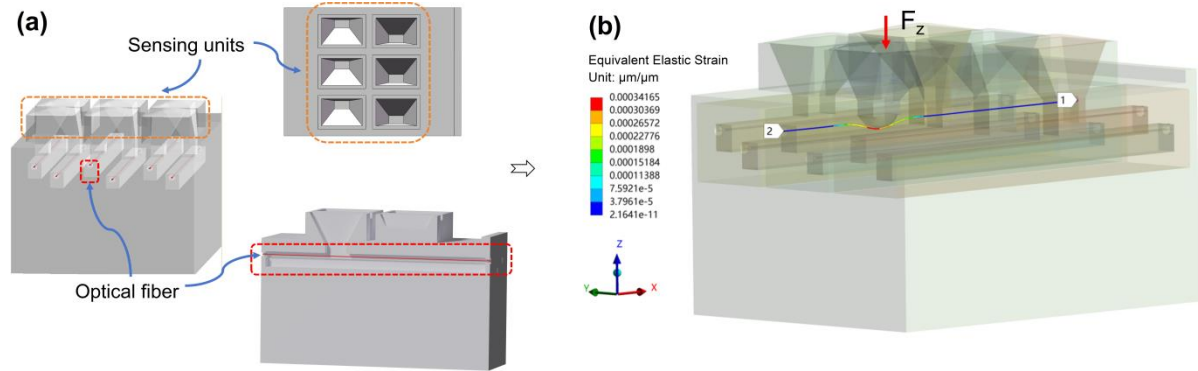


Fig. 3. Sensor unit structural details: (a) Optical fiber aggregation structure and sensing unit. (b) Equivalent elastic strain of PDMS and optical fiber based on simulation.

Fig. 3(a) presents the internal structure of the sensor. The area in the inverted tetrahedral shape and the lower fiber slot in each sensing unit is filled with PDMS (polydimethylsiloxane), which improves the spatial utilization efficiency of optical fiber arrangement and further concentrates stress. PDMS enables efficient force transfer from the contact surface to the embedded fibers, ensuring that strain is directly localized on the fibers, a design feature that enhances measurement repeatability. Its elasticity enables the fiber to quickly return to its original state upon unloading. In fiber optic force sensors, the optical fiber is typically encapsulated in an elastic

material to transmit external forces and isolate environmental interference. However, without rigid support, the elastomer often exhibits slow recovery after unloading, leading to hysteresis. To mitigate this issue, the sensor incorporates a rigid housing made of PLA, offering high stiffness for structural integrity and low creep for enhanced measurement stability.

Table I. MATERIAL PARAMETERS FOR FINITE ELEMENT SIMULATION OF SENSOR COMPONENTS

Material	Young's modulus (MPa)	Poisson ratio	Density (kg/m ³)
Polylactic Acid (Sensor Housing)	2553	0.35	1240
Silica (Optical fiber)	72000	0.17	2500
Polydimethylsiloxane (filler material)	1.56	0.49	1030

Fig. 3(b) illustrates the finite element analysis of the equivalent elastic strain on the optical fiber under a 10 N normal load, with simulation parameters detailed in Table I. The results indicate that strain is concentrated on the fiber segment directly beneath the sensing unit, peaking at approximately $341 \mu\epsilon$. While this micro-strain level is insufficient to induce a detectable change in total light intensity, the speckle pattern, due to its high sensitivity, undergoes a significant redistribution that is clearly captured by the camera. These analyses validate the sensor's design objectives and underscore the necessity of employing speckle pattern detection for high-performance tactile sensing.

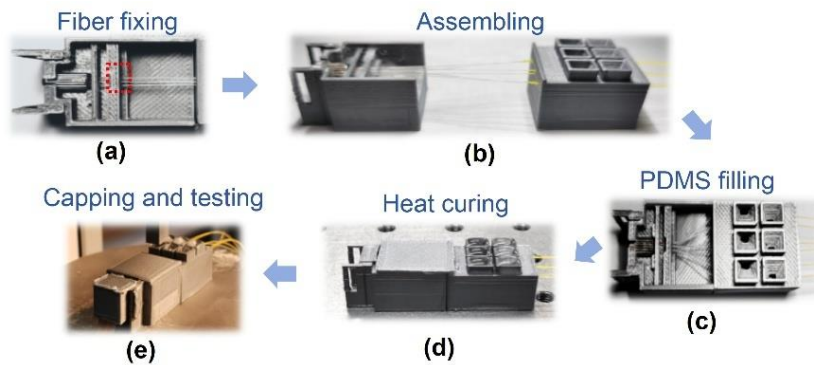


Fig. 4. Manufacturing flow chart of the sensor: (a) Fiber bonding. (b,c) Structural assembly. (d) PDMS pouring and curing. (e) Sensor packaging and testing.

Fig. 4 presents the fabrication process of the sensor. The fibers are first threaded through the micro-holes as shown in Fig. 4(a) into the lens interior. A camera captures the specklegram in real time, and the fiber position is adjusted until the fiber end is coaxially aligned with the micro-lens, enabling the capture of a complete specklegram. Then, the six fibers are immobilized in the micro-holes with a fast-curing adhesive. After that, the sensor module is systematically assembled as in Fig. 4(b). Mix PDMS with the base and curing agent in a 10:1 ratio, then inject the mixture into the sensor interior as in Fig. 4(c) to encapsulate the fibers. The sensor is then placed in a 50°C constant-temperature oven for heating and curing, ultimately forming a flexible contact surface on each sensing unit with cushioning properties.

3. Force Sensing Algorithms

The working principle of FSS is to detect changes in the measurand by analyzing variations in the output

fiber specklegrams. Specklegram processing algorithms can be broadly categorized into three types: feature-matching methods (e.g., ZNCC, Zero-Mean Normalized Cross-Correlation), machine learning-based pattern recognition [31], and deep learning-based image processing.

In this section, the performance of the sensing array is evaluated using ZNCC analysis, with a focus on key metrics such as force-response linearity, inter-unit crosstalk, and long-term stability. The insights gained from the ZNCC analysis provide guidance for the development of the DDL model presented in the following section.

The experimental setup of contact force sensing is illustrated in Fig. 5. The light source employs a 532 nm SMF-coupled laser, from which the light is uniformly distributed to the six sensor units through a 1×6 fiber splitter. The $8 \times 6 \text{ mm}^2$ probe of the force meter sequentially applies normal loads to each sensor unit fixed on the press platform at a constant rate. A miniature camera (resolution: 640×480 pixels, frame rate: 7.5 fps) captures the output specklegram from all sensor units in real time.

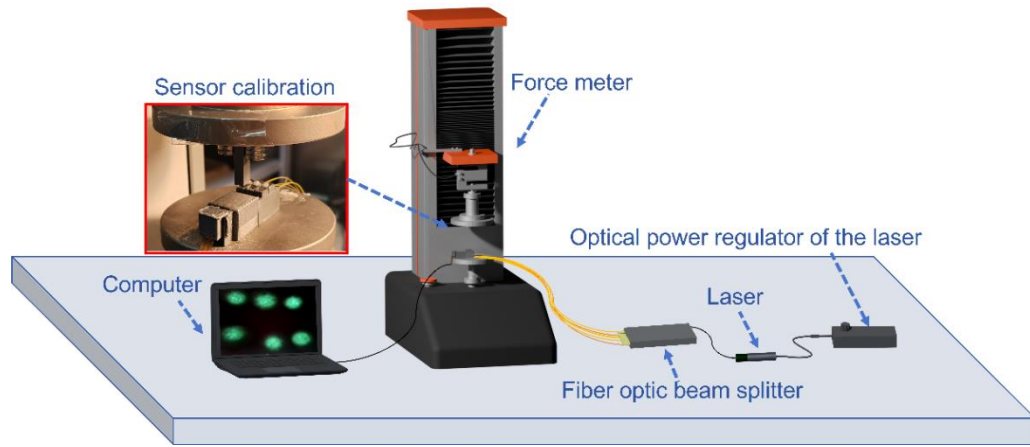


Fig. 5. Schematic diagram of the contact force sensing experimental setup.

3.1 Algorithms of ZNCC

The feature-matching method based on ZNCC enables force sensing by computing the similarity difference between the current and the reference specklegram as Eq. (3), which effectively suppresses the influence of light intensity fluctuations, demonstrating remarkable robustness [32].

$$\text{ZNCC}(I_0, I) = \frac{\sum_{i=1}^n (I_{0,i} - \bar{I}_0)(I_i - \bar{I})}{\sqrt{\sum_{i=1}^n (I_{0,i} - \bar{I}_0)^2 \sum_{i=1}^n (I_i - \bar{I})^2}} \quad (3)$$

where $I_0(i)$ and $I(i)$ represent the intensity values of the i -th pixel in the reference and captured speckle patterns, respectively, and n denotes the total number of pixels involved in the calculation. ZNCC evaluates the similarity between two patterns by comparing pixel-wise intensity variations after subtracting their mean values. Its value ranges from -1 to 1 , with higher values indicating greater similarity. Consequently, as the applied force increases, the speckle pattern deviates further from the reference state, causing the ZNCC value to decrease. Thus, ZNCC serves as a metric linking speckle pattern variations to the applied force on the FSS.

3.2 ZNCC-based Force Sensing Performance

To characterize the force sensing performance of the proposed sensor, each sensing unit underwent a complete force loading-unloading cycle. This process identified the linear operating range and established a linear fit between the applied force and ZNCC values. All six sensing units are sequentially tested, capturing

corresponding speckle patterns at various forces. After dividing the specklegram into six sub-specklegrams through edge detection, the ZNCCs for each unit are calculated as shown in Fig. 6, which show a linear dependence on the applied force. The loading/unloading curves exhibit remarkably small hysteresis.

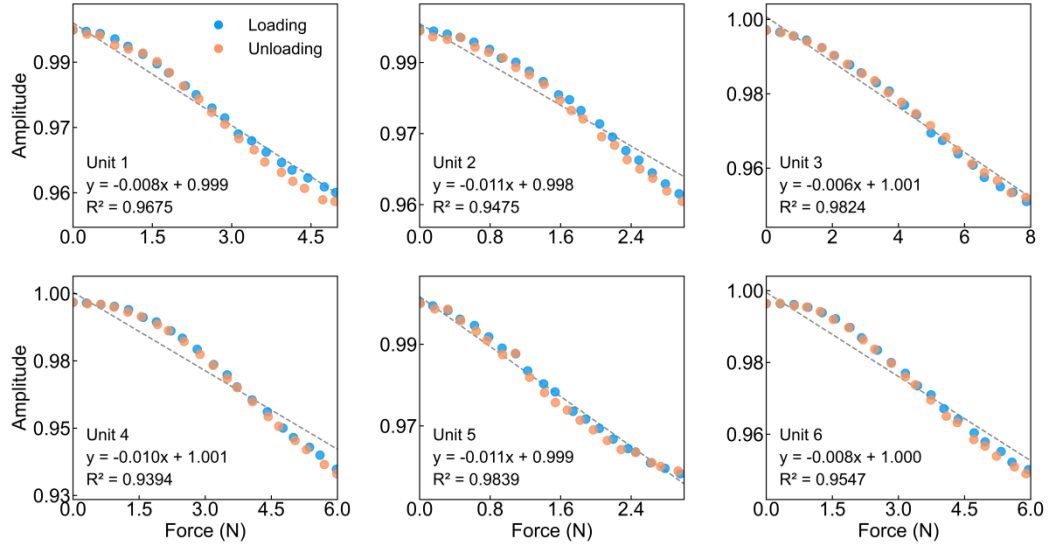


Fig. 6. Calculation results and linear fitting of ZNCC for units 1-6 under different contact forces.

To analyze the crosstalk between sensing units, force is applied on one of the units and the ZNCC change of all units was recorded. As illustrated in Fig. 7, inter-unit crosstalk is negligible, with ZNCC fluctuations in non-active units remaining indistinguishable from the system's background noise. This confirming the validity and effectiveness of the PLA rigid housing. This indicates that the proposed sensor array, based on ZNCC analysis of specklegrams, not only exhibits force-response linearity but also maintains independence of individual units, thereby establishing the necessary conditions for decoupling multi-point forces.

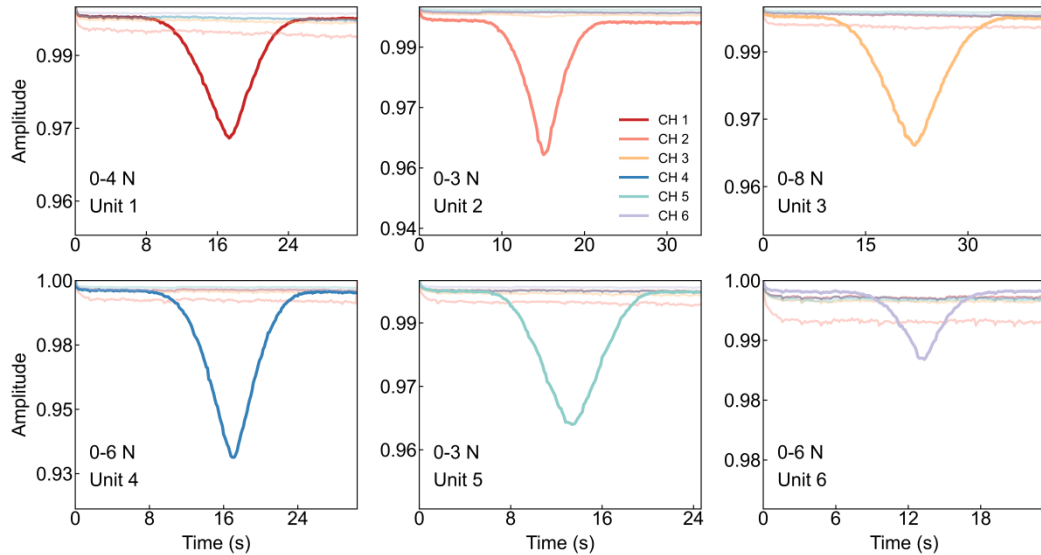


Fig. 7. Crosstalk curves based on ZNCC between units 1-6.

To evaluate the stability of the sensor, Unit 4 was tested continuously for four days. This unit showed the largest linear range in ZNCC-force curve. The first unloaded specklegram collected each day was used as the reference for that day, as shown in Fig. 8(a). The specklegrams collected over the four days exhibited minor

differences. To minimize the impact of specklegram brightness on the range of ZNCC, the average gray value of the specklegrams was kept similar during data collection. Data analysis indicates (Fig. 8(b)) that within 0-6 N range, despite minor errors introduced by variations in specklegram morphology and brightness fluctuations, the ZNCC vs force curve exhibits good repeatability.

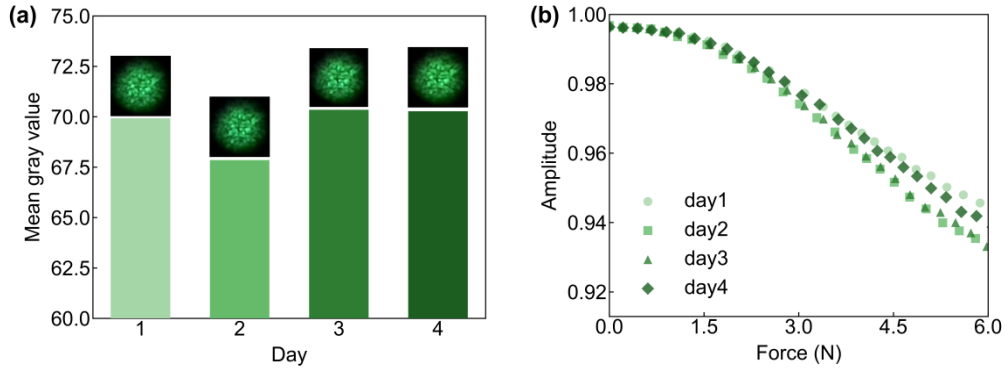


Fig. 8. (a) mean gray value of the reference specklegram for the 4-day dataset. (b) ZNCC calculated for each group of data vs. Force.

3.3 Practical Limitations of the ZNCC Analysis

ZNCC-based sensing methods are sensitive to variations in specklegram brightness and camera movement, which can affect measurement accuracy. For practical applications, it is highly important for the sensor to demonstrate interference resistance. While stable light sources can maintain constant output intensity, environmental factors may affect the optical path, causing variations in specklegram intensity. Consequently, the specklegram brightness differs each time the sensor is used.

Therefore, specklegram brightness still remains a factor affecting measurement accuracy even when ZNCC is used to analyze the specklegram. To quantitatively assess effects of brightness, seven specklegram datasets with different average gray values as shown in Fig. 9(a) were collected by adjusting the laser launching power. Average grayscale is calculated using pixels with values ≥ 15 within the specklegram region to reduce errors related to image size. Results (Fig. 9(b)) reveal that while ZNCC changes linearly with force, its proportionality coefficient varies with brightness.

Specifically, higher light intensity leads to smaller ZNCC variations under the same load, resulting in a flatter response curve. Notably, this reflects a shift in the sensitivity of the ZNCC metric rather than a change in the sensor's intrinsic mechanical properties. Consequently, accurate force prediction requires maintaining brightness levels consistent with the calibration state, or implementing an intensity compensation mechanism.

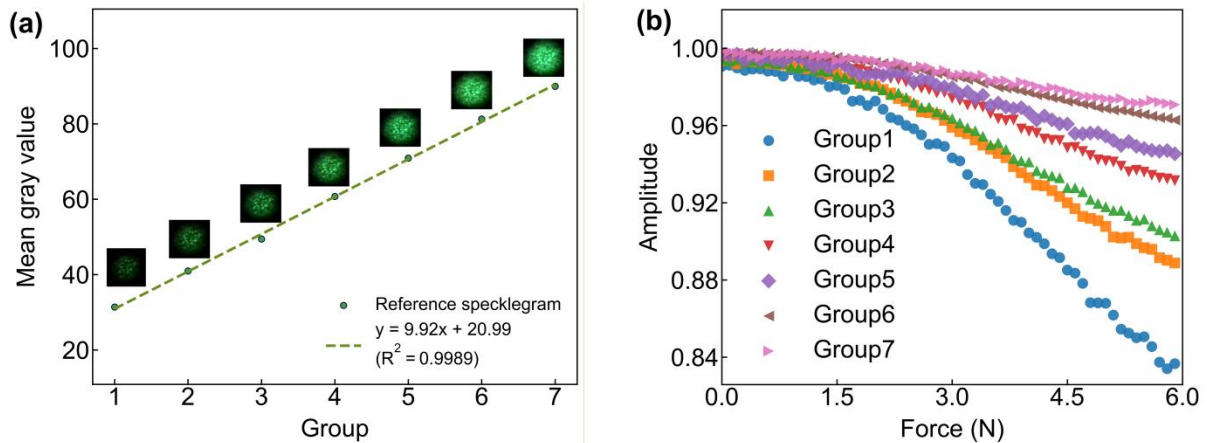


Fig. 9. Effects of specklegram brightness on ZNCC measurements. (a) Average gray value curve of the reference specklegrams for 7 sets of data (b) ZNCC vs force curves at different brightness levels.

Another noteworthy situation is that changes in the relative position between the optical fiber end face and the camera can affect sensor measurements, even if the input power of the sensor remains constant. Since ZNCC relies on pixel-wise correlation, changes in the relative position of specklegram introduce measurement errors, highlighting a critical limitation regarding the practical applicability of the current ZNCC algorithm.

4. Force Sensing Based on Deep Learning

In the previous section, we analyzed the practical limitations of ZNCC methods in specklegram sensing. Compared with ZNCC, data-driven neural networks offer significant capabilities of feature learning and extraction in highly complex data, which may potentially provide better performance than ZNCC in the specklegram analysis. However, existing research indicates that while CNN-based methods can improve sensing performance, they still face challenges in model generalization. Ref. [27-28] reports that when there is a certain time interval between test data and training data, long-term drift in specklegram morphology can cause a notable decrease in prediction accuracy of the CNN model.

As shown in Fig. 8(b), despite long-term drift in specklegram morphology, ZNCC variations induced by force exhibit a certain degree of temporal stability by using the newly captured image of each day as the reference. From a statistical perspective, this suggests that specklegram can still exhibit certain stability. Leveraging the differential logic of ZNCC, we propose a Dynamic Difference Learning (DDL) framework to evaluate its capacity for enhanced model generalization compared to single-image learning approaches. In the DDL network model, the learning objective is changed from “static features of specklegram under different forces” to “differential features of the specklegram induced by force.” The model takes both original specklegram (as reference benchmark) and specklegrams with force loading as inputs, with the corresponding forces used as supervision labels. Moreover, by updating the reference specklegram across different training datasets, the effect of morphological drift on model generalization can be reduced, thereby the prediction accuracy and robustness of the model are improved.

4.1 Neural Network Model Design

A dual-branch encoder architecture was designed to implement the proposed DDL model. The main task of the DDL model is to extract force-related features from the differences between two specklegrams. This design, which updates the reference specklegram, helps the model mitigate the effects of morphological variations unrelated to force, as shown in Fig. 10.

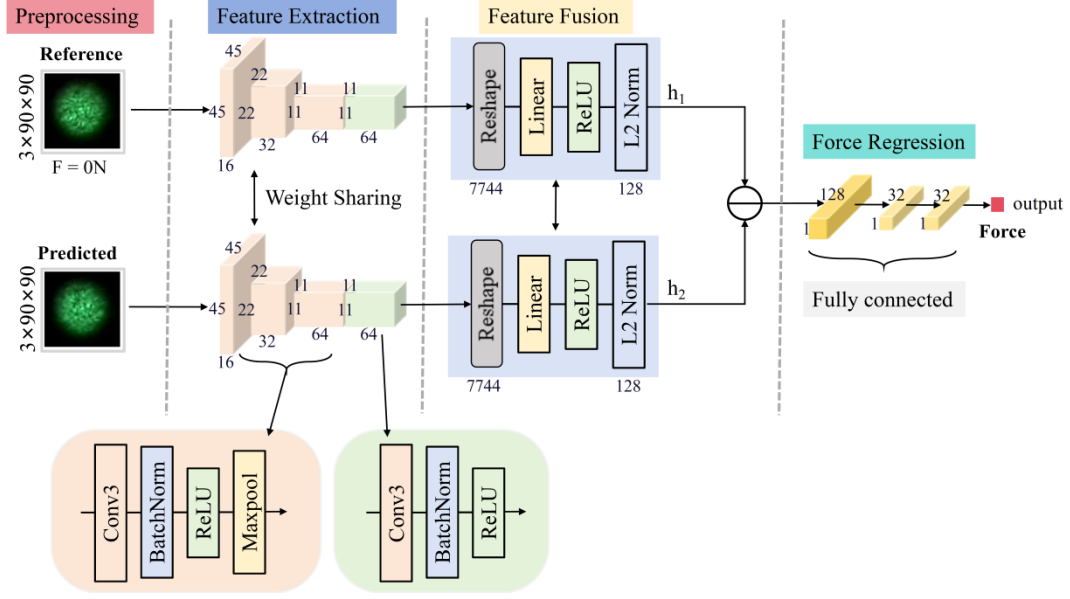


Fig. 10. Network architecture diagram of the DDL model. Abbreviations: Conv3 (3×3 Convolution Layer), ReLU (Rectified Linear Unit Layer), BatchNorm (Batch Normalization Layer), MaxPool (Max Pooling Layer), Linear (Fully Connected Layer), L2 Norm (L2 Normalization Layer).

The proposed model employs a dual-branch CNN architecture with shared weights to process paired $3 \times 90 \times 90$ specklegrams (reference and specklegrams with loading), where each branch utilizes a four-layer CNN encoder (Convolution + BatchNorm + ReLU) followed by three max pooling operations to extract 11×11 feature maps. These feature maps are first flattened and compressed through a linear layer, after which L2 normalization is applied. The feature differences between the two branches are then computed and fed into three fully connected layers to predict the force. Through this process, the nonlinear transformation from the high-dimensional feature space to the physical quantities is completed, and the end-to-end physical quantity reconstruction is ultimately achieved.

The objective of the model is to predict the force loaded on the sensor through dynamic difference analysis of specklegram images, which is a regression task. The mean squared error (MSE) is used as the loss function, while the root mean squared error (RMSE) and coefficient of determination R^2 (R-squared) are used as performance evaluation metrics, as shown in Eq. (4)–(6).

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (f_i - \hat{f}_i)^2 \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (f_i - \hat{f}_i)^2} \quad (5)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (f_i - \bar{f})^2}{\sum_{i=1}^n (f_i - \hat{f}_i)^2} \quad (6)$$

Where n is the sample size, f_i is the true value, $\hat{f}_i$ is the predicted value, and $\bar{f}$ is the average of the true values. MSE and RMSE measure the magnitude of prediction errors. R^2 evaluates the model's ability to explain the variance of the target variable, with values closer to 1 indicating better fit.

The model is trained using the Adam optimizer with an initial learning rate of 10^{-3} , and a learning rate scheduler reduces the rate by 50% every 10 epochs. Experimental results show that this configuration yields

optimal performance for the proposed task. Experiments are implemented using PyTorch 2.4.0 and accelerated with CUDA on an NVIDIA GeForce RTX 4060 GPU. To evaluate its suitability for real-time applications, we benchmarked DDL-Net on an NVIDIA RTX 4060. Using a batch size of 1 to emulate real-time data streaming, the model achieves an average inference latency of 0.88 ms.

4.2 Construction of the Specklegram Dataset: Composition and Preprocessing

4.2.1 Dataset Composition

A dataset was constructed through dynamic force loading to simulate continuous force changes in actual applications. To ensure consistency across array units, the force measurement range is uniformly defined as 0-10 N. A high-precision force meter was used to apply a cyclic loading of 0-10N at a step rate of 2 mm/min, while specklegrams were collected synchronously at 7.5 fps. While ZNCC is limited by saturation [33], which means the same ZNCC value can correspond to different specklegrams, deep learning can leverage additional features to achieve a larger effective measurement range.

To increase the diversity of the dataset and improve the model's robustness, data collection was conducted by changing two conditions: time periods and specklegram brightness levels. A time-segmented strategy was adopted to enhance the model's ability to capture dynamic features. For each sensor unit, data were collected on two separate days to induce natural speckle pattern variations, forcing the model to focus on force-induced features rather than static patterns.

To improve the model's robustness to brightness variations, four datasets were collected daily under distinct brightness levels with the average gray-scale value set at an interval of 20. Each set contained five complete force-loading cycles, with brightness kept constant within each set. At the beginning of each set, a 0-1N load was applied as a synchronization marker to ensure precise alignment between image sequences and force data. For each sensor unit, data were collected over two days, covering four brightness levels per day, yielding a total of eight sets.

The composition of the dataset is summarized in Table II. Data from the first two days were split into training and validation sets (8:2), with each set containing approximately 2,000 images, resulting in 16,000 images per sensor unit. The test set was collected on the third day, 24 hours after the training data acquisition, to rigorously evaluate the model's generalization to novel speckle patterns. The test set was collected 24 hours after the training data, consisting of three unseen brightness levels (mean grayscale values of 40, 60, and 80). For efficiency and to keep the test set at an appropriate size, the number of loading cycles per brightness level was reduced from five to two. On the same day, an additional test set (test2) was collected to further evaluate the model's robustness to brightness variations. This complete data collection procedure was replicated across all six sensing units. In both the training and testing phases, the first frame of each loading sequence served as the fixed reference image.

Table II. Dataset composition

Subset	Force range	Loading cycles	Brightness levels	Collection time	Images per unit
Train/Val	0-10N	5 per set	4 per day	Day 1-2	~16,000(split8:2)
Test 1	0-10N	2 per set	3 unseen (40, 60, 80)	Day 3	~2,600
Test 2	0-10N	2 per set	20-100, step 10 (9 levels)	Day 3	~7,200 (Unit4)

This approach offers a new strategy for practical real-time applications. Similar to the 'zeroing' operation in standard force sensors, the reference image can be updated when system drift is detected or the sensor returns to

an unloaded state. This mechanism resets the sensing baseline.

4.2.2 Specklegram Preprocessing

The specklegram preprocessing method is illustrated in Fig. 11. First, the captured video is decomposed into individual frames. Next, the Canny edge detection algorithm is applied to locate and segment six independent specklegram regions. To preserve the original morphological features, zero-padding is used to standardize each specklegram to 90×90 pixels, thereby avoiding feature distortion from image deformation.

Meanwhile, the force data recorded during force loading at a sampling rate of 7.5 Hz is synchronized matched with the image frames to establish a force-image correspondence. For each dataset, the reference specklegram is defined as the first acquired image under the 0 N loading condition in chronological order.

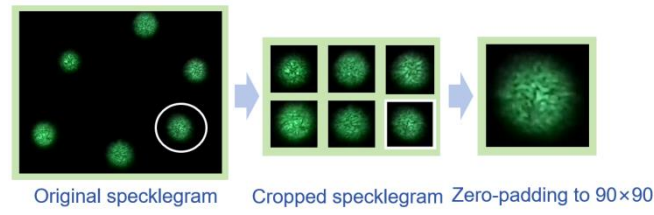


Fig. 11. Schematic diagram of specklegram preprocessing.

Specifically, in order to simulate the pixel-level misalignments in specklegrams caused by edge detection segmentation, random shift augmentation during training is introduced, in which the input image is randomly translated within a small pixel-level range along the x and y directions and merged with the original data. This strategy also mitigates the effect of minor displacements in real scenarios (e.g., camera shake), encouraging the model to learn feature representations that are robust to positional shifts, thereby enhancing generalization during testing.

5. Experimental Results and Discussion

5.1 DDL Model Performance and Generalization

5.1.1 Training of DDL Model

As shown in Fig. 12(a), using Unit 4 as an illustrative example, the training process exhibits stable convergence with both training and validation losses decreasing rapidly.

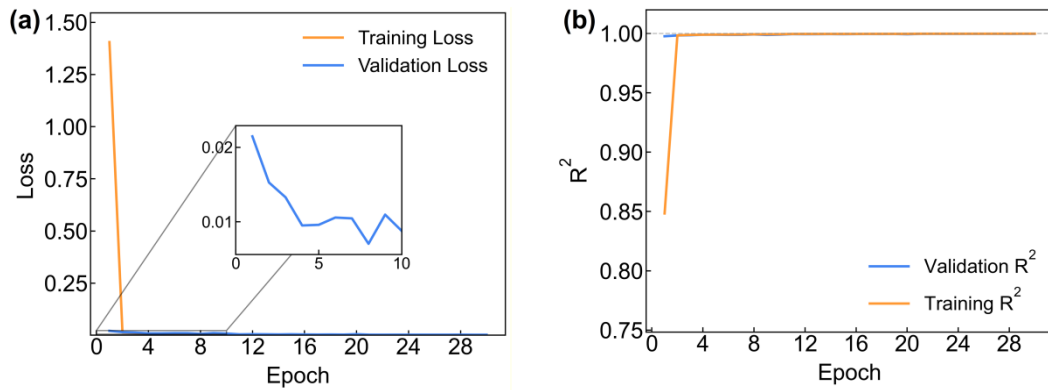


Fig. 12. (a) Loss curve during network training. (b) R^2 curve during network training.

This demonstrates that the proposed DDL model successfully learns discriminative specklegram features.

Furthermore, the model achieves reliable predictive accuracy on the validation dataset, as indicated by the R^2 value approaching 1 over training epochs (Fig. 12(b)), demonstrating agreement between predicted values and the ground truth. The overall predictive performance across sensor units is summarized in Table III, further confirming the robustness and reliability of the proposed method.

Table III. Comprehensive performance evaluation across training and validation sets

Unit id	Training Set		Validation Set	
	R^2	RMSE	R^2	RMSE
1	0.9998	0.0319	0.9998	0.0422
2	0.9999	0.0253	0.9998	0.0390
3	0.9998	0.0426	0.9995	0.0614
4	0.9998	0.0359	0.9997	0.0524
5	0.9999	0.0264	0.9997	0.0448
6	0.9997	0.0475	0.9988	0.1034

5.1.2 Model Generalization Analysis

Deep learning in specklegram sensing faces a challenge: model performance degradation, indicated by reduced accuracy on test data collected after a time interval. This may arise from: (1) speckle pattern variations resulting from disturbances along the entire optical path compounded by environmental fluctuations; (2) long-term sensor degradation. The former compromises accuracy; the latter limits predictive capability, as unstable sensors hinder generalization..

To demonstrate the DDL model's effectiveness in mitigating the aforementioned issue, we evaluated it on the previously described test set, thereby enabling a rigorous examination of its generalization performance under both speckle pattern drift and illumination variations. To provide a direct performance benchmark, we constructed a baseline model by removing the reference branch and difference operation, resulting in a single-specklegram prediction network with a similar architecture and parameter count. Both models were trained on the same dataset and evaluated on the same test set to ensure a fair and consistent comparison.

Table IV. Generalization ability comparison on Test Set 1 (Table 1): DDL versus single-image training

Method	DDL (Ours)		Single-specklegram	
Unit id	R^2	RMSE	R^2	RMSE
1	0.9746	0.4819	0.9636	0.5766
2	0.9958	0.1926	0.9950	0.2110
3	0.9930	0.2480	0.9861	0.3487
4	0.9991	0.0910	0.9987	0.1081
5	0.9975	0.1494	0.9970	0.1633
6	0.9976	0.1444	0.9961	0.1854

Table IV shows the prediction results of different sensor units evaluated on test sets collected under identical conditions. Across units, the DDL achieves lower RMSE values and higher R^2 , demonstrating better generalization. This improvement can be attributed to the model's ability to mitigate the impact of specklegram drift by dynamically updating the reference specklegrams.

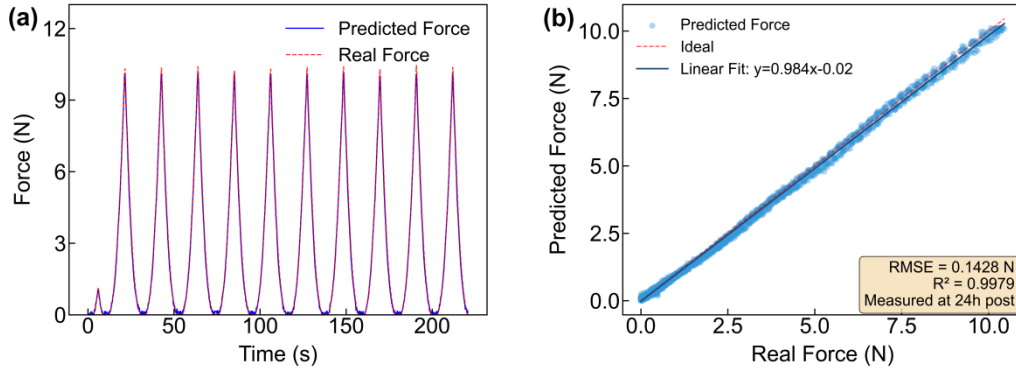


Fig. 13. Force prediction for Unit 4 under cyclic loading (mean grayscale = 60; test data 24 h after training). (a) Predicted and actual force over 10 cycles. (b) Correlation between predicted and real forces.

Fig. 13 shows the predicted force closely following the real force over ten 0-10 N loading-unloading cycles using test data collected 24 h after training, validating the sensor's dynamic sensing capability. The predicted and real forces exhibit a high correlation close to the ideal $y = x$ line, indicating high accuracy and consistent performance.

It is observed that model performance can be influenced when the brightness of the specklegram during testing differs from that in training. To systematically evaluate generalization under varying brightness conditions, we employed the brightness-variation test set of Unit 4 (Table II-Test 2), which includes nine different mean grayscale levels.

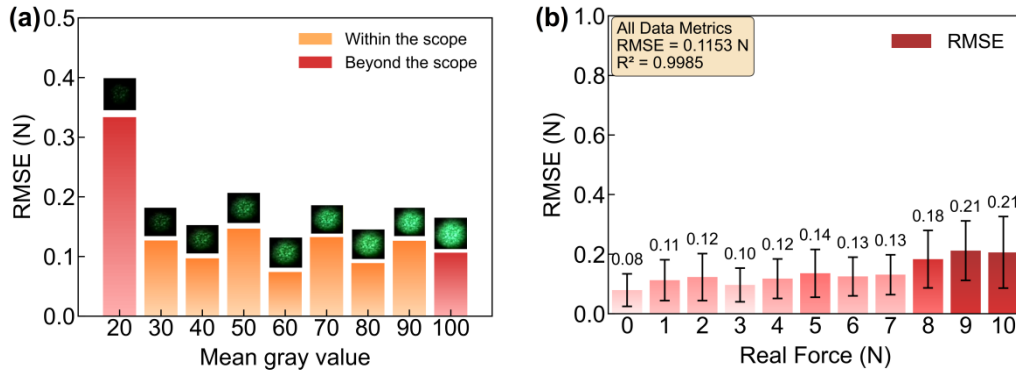


Fig. 14. Performance metrics on Test 2 dataset: (a) Prediction RMSE versus light intensity.(b) Error distribution of averaged predictions across specklegram grayscale values (range from 30 to 100).

Fig. 14(a) evaluates the model's generalization to intensity variations via RMSE across dataset groups, including reference specklegrams. Reliable predictions are even achieved at a mean grayscale value of 100, which is outside the training range. However, at a mean grayscale value of 20, the prediction error increases notably due to pixel loss in overly dim specklegrams.

Fig. 14(b) shows RMSE versus applied force (0-10 N in 1 N steps) over the 30-100 grayscale range. The model achieves an overall RMSE of 0.1153 N and $R^2 = 0.9985$ on test data collected 24h after training, demonstrating robustness to intensity variations and suitability for real-time sensing. These results indicate that the proposed method is robust and stable, exhibiting minimal impact from intensity variations, and highlight its suitability for real-time sensing scenarios.

Based on this analysis, a key advantage of the proposed scheme is that the DDL framework leverages

feature discrepancies between paired specklegrams to decouple intrinsic signal from environmental interference. By dynamically updating the reference specklegram similar to a zeroing operation, the framework effectively mitigates the impact of environmental variations on prediction accuracy.

5.2 Force Prediction by Sensor Array

The distinctive advantage of the designed sensor array lies in the independence of its units and their ability to sense spatial stress distribution, which enables synchronous measurement of both multi-point stress distribution and the resultant force. To validate this capability, simultaneous 0-20 N loading-unloading tests were performed on all units using a force meter, with each unit's contact force being predicted by its specifically calibrated model. Let F_{total} denote the predicted resultant force from the sensor array, and F_i is the predicted force from the i -th sensor unit. Under ideal conditions, they should satisfy the following relationship (Eq. (7)):

$$F_{\text{total}} = \sum_{i=1}^6 F_i \quad (7)$$

A dynamic force of 0–20 N was applied to the sensor array, and the trained network model was used to predict the force at each unit. As shown in Fig. 15, due to height discrepancies among the sensor units, their contact timings with the force probe varied—this led to a temporal offset in the initial points of the force curves corresponding to each unit.

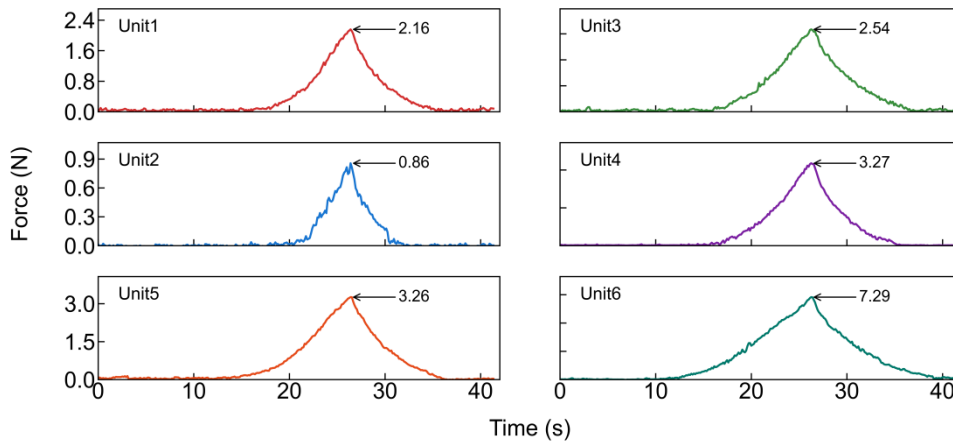


Fig. 15. Predicted force variation curves over time for Units 1–6.

The predicted forces from individual sensor units were summed and compared with the reference values from the force meter as shown in Fig. 16. The results demonstrate consistent agreement between predicted and reference values, with an RMSE of 0.4206 N, accounting for approximately 2.05% of the full-scale range. Although some prediction errors were observed, the overall trend closely matches the reference values, thereby validating the sensor array's capability to simultaneously sense force distribution and resultant force.

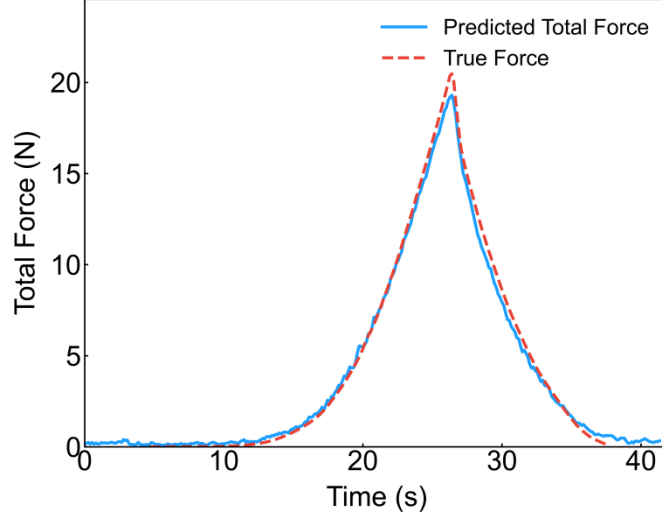


Fig. 16. Comparison between predicted resultant force and actual force.

6. Comparison with Other Tactile Sensing Technologies

Table V compares the proposed sensor with current fiber-optic tactile sensing studies, which highlight its advantages in hardware design and algorithmic performance.

The proposed 2×3 sensor array enables independent multi-point force sensing and resultant force estimation, while its adoption of MMF, PDMS, and USB cameras enables reliable sensing with lower cost and compact integration suitable for robotic and prosthetic applications. Although our multi-MMF design incurs a slight hardware cost increase compared to single-MMF distributed sensing, it is more practical as it bypasses the substantial dataset costs and complex signal decoupling inherent to simultaneous multi-point sensing. Moreover, the speckle-based approach is well-suited for this structure, offering lower demodulation costs than FBG sensors and high sensitivity compared to intensity-based sensors. The dual-branch DDL-Net improves accuracy by reducing RMSE across units through updated specklegram reference processing, which functions as a 'zeroing' operation for practical applications, while adding negligible computational overhead, requiring no enlargement of the dataset, and supporting real-time embedded sensing.

Table V. Performance comparison of fiber-optic tactile sensing technologies

Reference	Sensing Method	Measurement Range	Array Size	Result
[34] (2023)	FBG + BP+ELM	0-3.5 N	2×3 array with four FBGs;Sensing area 20×30 mm ²	MRE < 4.2%
[35](2023)	CFBG + Res-VGG-Net	2-10 N	40×40×2.5 mm ³ sensing area with two CFBGs	RMSE ~ 0.08 N
[36] (2023)	MMF specklegram + Res-Net	0-0.0368N	Single fiber-end probe	MAE < 2%
[37] (2023)	MMF specklegram + VGG-Net	2-3N(0.5N intervals)	Single reflective fiber probe with 9 contact positions	Classification accuracy 100% (for 9 positions, 3 force levels)

This work	MMF specklegram + DDL-Net	0-10 N (single unit), 0-20 N (total force)	2×3 array with 6 fibers; Sensing area 15×17 mm ² ; Unit Sensing Area 5×7 mm ²	Single unit: $R^2 = 0.9985$, RMSE = 0.1153 N; Total force: RMSE = 0.4206 N, 2.05% of full range
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7. Conclusion

This paper presents a miniaturized tactile sensor array (2×3 units) based on MMF specklegram detection. Each independent unit enables simultaneous multi-point force sensing and resultant force estimation with high stability and repeatability.

This study investigates the practicality of existing speckle sensing algorithms and specifically addresses the limitations of the ZNCC method, including susceptibility to light intensity fluctuations, camera displacement and saturation effects. Building on the computational logic of ZNCC, we propose an DDL network that improves accuracy and reduces RMSE across units through updated reference specklegram processing. By constructing an enhanced dataset incorporating varied light intensities and data collected at multiple time points, the model's generalization capability is improved. Functionally, The proposed DDL-Net acts as both a filter and a function fitter. Compared to the global calculation of ZNCC, the DDL-Net selects features more adaptively, thereby avoiding the shortcomings of ZNCC methods. it overcomes the issues of inconsistent linearity and limited measurement range inherent to ZNCC. Furthermore, its robustness to light intensity variations simplifies practical deployment by eliminating the need for strict laser power tuning, requiring only sufficient brightness for operation.

Experimental results demonstrate the sensor units exhibit high measurement accuracy within the range of 0–10 N. The DDL network maintains consistent generalization over the test set obtained 24 hours after the training set and robustness to light intensity variations, which is achieved via updating reference specklegram. Furthermore, the sensor array enables independent multi-point force measurement, with predicted forces closely matching references (RMSE = 0.4206 N, 2.05% of full range). By learning the differential features between the current specklegram and the reference, rather than only the features of the current specklegram, the network can better track force-induced changes in the specklegram, thereby more accurately reflecting the applied force on the sensor, which enhances interpretability and demonstrates the feasibility of specklegram-based sensing.

For fiber optic tactile sensing, this study investigates the practicality of speckle sensing and provides a technical solution that integrates low cost, high accuracy, and an sufficient measurement range, thereby exhibiting potential for deployment in fields such as robotic precision manipulation and intelligent prosthetics.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Fei Deng: Conceptualization, Methodology, Software, Data Curation, Writing - Original Draft. **Yan Liu:** Conceptualization, Writing - Review & Editing, Project administration, Funding acquisition. **Zhibo Hou:** Validation, Data Curation, Formal analysis. **Ziling Wang:** Investigation, Data Curation, Formal analysis. **Zhengying Han:** Data Curation, Formal analysis. **Chengcheng Hou:** Data Curation, Formal analysis.

Muguang Wang: Conceptualization, Resources, Supervision. **Fengping Yan:** Conceptualization, Resources, Supervision.

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: